



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT (PART II)

Date: 27 July 2026

In addition to Part-I (General Handout for all courses appended to the timetable), this portion gives further details regarding the course.

Course No.: CHEM U110
Course Title: CHEMISTRY LABORATORY
Instructor-in-charge: BIBHAS RANJAN SARKAR

Instructors:, Prof. Bibhas R. Sarkar, Prof. Ram K. Roy, Prof. Inamur R. Laskar, Prof. Pritam K. Jana, Prof. Anish Rao, Ankit Payal, Ankita Narain, Pinki Kanodia, Ritu Kumari, Sakshi Kushwaha, Prashant Mani Tripathi, Ayesha Rahman, Sakshi Bajaj, Pinki, Parmeshthi Parikh, Rohit Sharma, Teena, Ritu Rani, Kamal Lochan, Afrisham Ahmad, Harshi Saxena, Priyamvada Singh, Aashima, Harsh Parashar, Ritik Pathania, Pragya Giri, Arun Pratap Singh, Priyanshi Choudhary, Shreya Mittal, Preeti Meena, Ankita Sharma, Mukesh Kumar, Tarun

1. AIMS AND LEARNING OBJECTIVE:

The main objective of this course is to educate the students about different aspects of chemistry experiments. The students will carry out experiments that will expose them to experimental methods and integrate theoretical knowledge and concept into practical experience. Students will also learn the operation of some scientific equipment for performing experiments.

2. TEXTBOOK:

Lab Manual and Lab Workbook for Chemistry Laboratory: Dept. of Chemistry, BITS Pilani, Pilani Campus

3. REFERENCE BOOK:

Vogel's textbook of quantitative chemical analysis, Prentice Hall, 2000.

4. COURSE PLAN:

The students will perform the following TEN experiments.

List of Practical Experiments (Evaluative)	
Exp. No.	Experiment Title
1.	Determination of total hardness of water with EDTA
2.	Estimation of copper by Iodometry
3.	Determination of the pH curve of an acid-base titration
4.	Synthesis of dibenzalacetone
5.	Colorimetric determination of unknown concentration of $K_2Cr_2O_7$ solution
6.	Dissociation constant of a weak electrolyte by Conductometry
7.	Validation of Heisenberg's Uncertainty Principle Using Single-Slit Diffraction
8.	Identification of some organic compounds



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Demonstration experiments (Non-evaluative)	
A	Mechanochemical synthesis of a Schiff's base ligand and its Copper(II) complex – a green chemistry approach
B	Polarimetry experiment (optional)

5. EVALUATION

Component (Marks)	Weightage %	Date and Time
Laboratory Work and Reports (168 M)*	84%	Continuous
Comprehensive Examination (32 M)	16%	To be announced
Total 200 M	100%	

*** Each Experiment carries 28 Marks. Best 6 out of 8 experiments will be considered for grading.**
Demonstration experiments are non-evaluative

6. IMPORTANT NOTES:

- ✓ Students must complete and submit the completed lab workbook on the same day of the lab class, within the same class duration.
- ✓ Students are expected to read the allotted experiments from the manual before coming to the lab.
- ✓ Students **MUST** bring their **OWN** Calculator, Laboratory Manual and Laboratory Workbook.
- ✓ Use of any electronic gadget such as mobile phones, tablets, laptops, smart watches, etc. is **STRICTLY** prohibited during the laboratory class.
- ✓ Students must obey the laboratory safety rules. Failure may result in denial of the entry to the lab.
- ✓ A **total of 08 experiments (04 in Cycle 1 and 04 in Cycle 02)**, each of 28 marks (Attendance 02, Performance 04, Cleanliness 02, and Lab record 20) will be conducted, out of which the **total score of best 06 experiments** (total 168 marks) will be considered for final evaluation (i.e., two buffers).
- ✓ There will be not any separate Mid-Semester Examination, scores of first four experiments will be considered for mid-semester exam grading.
- ✓ A Comprehensive Quiz Examination (32 marks) will be held at the end of the semester. This will include questions from all 8 experiments.
- ✓ **Minimum performance requirement for a valid grade:** To clear the course, a student should obtain at least 30 % of the average marks of the top 10 performers in the class, or 40 % of the median (whichever is lower), of the course. **If any student gets marks lower than the prescribed standard mentioned above, he/ she may be awarded NC grade.**

7. MAKE-UP:

Please also refer to Part I of the handout given in the timetable. Make-up for lab class is highly discouraged, as sufficient buffers are provided. Make-up for the comprehensive exam may be considered for genuine cases as per the Institute's rules **against application** by the student supported by valid documentation. The students are **ADVISED NOT TO MISS ANY LABORATORY CLASS**.

8. NOTICE: Notices concerning this course will only be displayed on the **NALANDA ONLY**. **Students are advised to be in-sync with their institutional e-mail for important information about the course.**

Instructor-In-Charge
CHEM U110